

Final assessment of the review of the DataMod 2025 article

Probabilistic Algorithm with Dynamic Load Balancing for GPU-Accelerated Tumor Growth Simulations

Reference document

August 2026

1 Context

The three original review reports have been compared with the version finally published in *Lecture Notes in Computer Science* (LNCS), vol. 16421, Springer Nature, 2026, pp. 36–53. The overall impression is that the review was considerably more thorough than it might appear at first glance: virtually all the criticisms raised by reviewers 1 and 2 were addressed, whilst the main remaining reservations are the most stringent ones raised by reviewer 3.

La referencia de la publicación es:

M. I. Capel and L. Rodríguez Domingo, “Probabilistic Algorithm with Dynamic Load Balancing for GPU-Accelerated Tumor Growth Simulations”, in L. Lestingi, G. Salaün and J. I. Requeno Jarabo (eds.), *From Data to Models and Back – DataMod 2025*, Lecture Notes in Computer Science, vol. 16421, Springer Nature, 2026, pp. 36–53. DOI: [10.1007/978-3-032-25552-5_3](https://doi.org/10.1007/978-3-032-25552-5_3).

2 Correspondence between revisions and the published version

Rev.	Original review	What was done in the published version	Status
R1	The introduction did not make it sufficiently clear <i>context</i> → <i>problem</i> → <i>gap</i> → <i>contribution</i> .	The published introduction follows this structure almost exactly: cost of CAs → scalability issues → synchronisation/load balancing → constraints imposed by previous work → a probabilistic GPU approach without costly global synchronisation.	Resolved
R1	Explain CUDA to non-specialist readers.	CUDA is explicitly defined as <i>Compute Unified Device Architecture</i> , a parallel computing platform developed by NVIDIA.	Resolved
R1	Minor editorial issues, such as the use of phrases like ‘numerous works’, inconsistencies between AC and CA, references, etc.	The LNCS version has been extensively rewritten, and these issues no longer appear as such.	Resolved
R1	The figure for the Moore neighbourhood is incorrect or inconsistent.	The layout of the form and its explanatory notes have been corrected.	Resolved

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Rev.	Original review	What was done in the published version	Status
R1	The CUDA listings disrupted the presentation and added little scientific value.	The final version reorganises and consolidates the description of the implementation, placing the emphasis on algorithmic and architectural decisions.	Resolved
R1	The title “Demonstration of the α -cell...” It didn’t really match the content.	The way the probabilities are expressed has been reorganised, and that problematic presentation has been removed.	Resolved
R1	Unclear baseline; it was not clear what was being compared or why.	The final evaluation provides a much clearer distinction between the sequential baseline, the speed-ups, the comparison with previous work and the analysis of the various GPU strategies.	Mostly resolved
R1	In contrast to the previous study [20], which was purely narrative and difficult to interpret.	The final version substantially expands the comparison and puts the performance into context relative to previous implementations. However, it is still not a controlled experiment in which both implementations are run on identical hardware.	Partially resolved
R1	Biological realism was not assessed: accelerating a model without knowing whether it produces reasonable results limits its contribution.	<i>Sensitivity Analysis</i> is now added, with 10 runs per configuration, variations of μ , α and ρ_{re} , and an analysis of the effect on the final tumour size and execution time. It also ties in with the underlying experimental study.	Partially resolved
R2	Details regarding the selection and calibration of biological parameters were missing.	The justification is expanded and an explicit analysis of the calibrated values is introduced.	Reasonably resolved
R2	<i>Sensitivity analysis</i> was missing.	A whole new subsection has been added, 5.2 <i>Sensitivity Analysis</i> , varying μ by $\pm 20\%$, α by $\pm 20\%$ and ρ_{re} by $\pm 10\%$, with 10 runs per configuration.	Clearly resolved
R2	Include a bibliography on <i>sensitivity analysis</i> in biological modelling.	The section explicitly includes methodological references on sensitivity and uncertainty analysis.	Resolved
R2	Analyse the performance overheads introduced by GPU implementation decisions.	Detailed profiling has been added. <code>transitionFunction()</code> accounts for approximately 81% of the execution time, and its occupancy drops to around 36% in later iterations due to the uneven distribution of the workload.	Resolved
R2	Atomic operations and synchronisation could become a bottleneck as the problem grows.	The architectural rationale becomes much clearer: the probabilistic redesign allows for thread-independent updates and avoids global synchronisation and extensive atomic operations.	Resolved
R2	The trade-off between probabilistic choices and performance was not quantified.	The <i>sensitivity analysis</i> shows noticeable changes in tumour behaviour, whilst the running time changes by less than 1.5%.	Resolved
R3	The parallelisation strategy lacked depth.	The published version covers profiling, occupancy, SM utilisation, memory and the 16 rows/columns, 64 regions and hybrid strategies.	Largely resolved
R3	It was unclear what the difference was between the novelty of the probabilistic model and that of the GPU implementation.	<i>Related Work</i> and <i>Introduction</i> They distinguish more clearly between these two aspects and explain that the methodological problem lies in making probabilistic CA compatible with scalable GPU execution.	Resolved

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Rev.	Original review	What was done in the published version	Status
R3	The GPU section was hard to follow; shared memory and unclear CUDA functions.	The final exhibition has been reorganised and now focuses much more on architectural decisions and measured performance.	Mostly resolved
R3	The details on parallelisation were scattered across sections 3 and 4.	The final version has a much more coherent structure and subsequently devotes section 5.3 <i>Scalability Analysis</i> to the effect of partitioning.	Resolved
R3	Main review: It is only really compared against CPU/sequential; a state-of-the-art (SOTA) GPU implementation is missing as a baseline.	GPU-related papers are included and discussed, such as Cagigas-Muñiz y Du et al., y The performance is quantified. However, a state-of-the-art (SOTA) GPU baseline is not reimplemented and run on the same hardware.	Pending
R3	If a direct comparison cannot be made, a state-of-the-art method should be re-implemented.	That experimental reimplementation is not being carried out <i>apples-to-apples</i> .	un resolved
R3	The fact that certain times were worse than those of [20] could cast doubt on its suitability for the GPU.	The additional cost of probabilistic rules is explained much more clearly, and it is argued that the performance is on a par with state-of-the-art (SOTA) CA GPUs despite offering greater biological expressiveness.	Partially resolved

All three reviewers started from a relatively favourable assessment: all three issued a *weak accept* recommendation. However, Reviewer 3 pinpointed precisely the most scientifically challenging issue: it was not enough simply to improve the explanation of the method; instead, they requested a new comparative experiment with another GPU implementation.

3 Balance global

If we limit the reports to the most significant scientific critiques, the overall picture can be summarised roughly as follows::

Resolved: ~75% **Partially: ~15%** **Pending: ~10%**

Apart from editorial corrections, the review yielded two new scientific findings that clearly strengthen the article.

3.1 Sensitivity analysis and the distinction between biological dynamics and computational cost

The inclusion of sensitivity analysis allows us to demonstrate experimentally that modifying biological parameters significantly alters the simulated dynamics without substantially increasing the computational cost. For example, varying ρ_{re} results in changes of approximately -14.8% and $+16.2\%$ in the final tumour size, whilst the running time changes by only around $+1.1\%$ and $+1.3\%$.

This result quite clearly distinguishes between two dimensions that appeared to be more intertwined in the initial version:

- **Biological dynamics:** sensitivity of tumour progression to the model parameters.
- **Computational performance:** relative stability of implementation costs in the face of such fluctuations.

3.2 Scaling and spatial origin of the imbalance

The second finding, and probably the most important from an HPC perspective, is the scalability analysis. The published version concludes by identifying that the problem is not simply a matter of ‘using CUDA’. With a partition into 16 large regions, a notable imbalance arises because tumour growth is spatially concentrated; a partition into 64 regions distributes the active cells more evenly amongst the SMs; and the hybrid strategy emerges precisely as a response to the trade-off between load balancing and memory locality.

Consequently, the article ceases to be merely a comparison of CPUs and GPUs and begins to reveal a more general architectural phenomenon: **the spatial concentration of activity within an irregular probabilistic workload leads to non-uniform utilisation of the Streaming Multiprocessors.**

4 Scientific continuity leading up to Euro-Par 2026

The main limitation that remains unaddressed in the article published in this book, *DataMod* (Springer Nature), coincides almost exactly with the natural starting point for further work towards Euro-Par 2026: to investigate more rigorously why GPU imbalance occurs, to measure it directly on the SMs, and to analyse partitioning strategies for spatially irregular probabilistic workloads.

In retrospect, the research career can be summarised as follows:

DataMod 2025: *Can we run this probabilistic CA efficiently on a GPU?*

⇒

Result: Yes, with speed-ups of up to approximately 219×, although there is some degradation in occupancy and load imbalance.

Euro-Par 2026: *Why does this imbalance arise, and how can we characterise and mitigate it?*

⇒

Working hypothesis: spatial hotspots + probabilistic irregularity → uneven distribution across machine learning models → need for 16/64/hybrid partitioning strategies and specific imbalance metrics.

This approach is scientifically more sound than presenting Euro-Par as a mere extension of the DataMod paper featuring ‘more experiments’. The subsequent work takes as its starting point an empirical limitation identified in the published paper and transforms it into a more general HPC question concerning **hotspot-induced imbalance in irregular probabilistic GPU workloads.**

5 Conclusion

The version published in Springer/LNCS clearly addresses most of the objections raised during the DataMod 2025 review. The most significant improvements were not merely editorial, but methodological and experimental: the contribution was clarified, the parallelisation strategy was strengthened, sensitivity analysis was added, the costs of the kernels were quantified, and the scalability of different partitioning schemes was explicitly investigated.

The main outstanding objection is the lack of a comparison *apples-to-apples* compared with a state-of-the-art (SOTA) GPU implementation running under identical experimental conditions. It is precisely for this reason that further work can be underpinned by a clear and legitimate rationale: to explore in greater depth the architectural characterisation of imbalance and load-balancing strategies for irregular probabilistic workloads.